MLS Lemma 2.3

(p. 28)

We wish to prove eq. (2.12):

$\hat{a}^2 = aa^T - ||a||^2 I$

In the code below,

$aa^T - ||a||^2 I$

is defined as RHS1, and $\hat{a}^2$ as LHS1. The equation will be proven if

LHS1 - RHS1 = 0.

```
syms a_1 a_2 a_3 real

a       = [a_1; a_2; a_3]
```

a =

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

```
hata    = hatm_sym(a)
```

hata =

$$\begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$$

```
% hat_sym is a function written by
%   NJ Theron, not made available to students, to create
%   appropriate skew-symmetric matrix for cross product
%   calculation, from the column vector passed to it as
%   argument.  Students are strongly encouraged to write
%   both a symbolics and a numerical function like this,
%   for their own libraries.

norma2 = a'*a % norm of vector a, raised to power 2
```

norma2 = $a_1^2 + a_2^2 + a_3^2$

```
LHS1 = hata*hata
```

LHS1 =

$$\begin{pmatrix} -a_2^2 - a_3^2 & a_1 a_2 & a_1 a_3 \\ a_1 a_2 & -a_1^2 - a_3^2 & a_2 a_3 \\ a_1 a_3 & a_2 a_3 & -a_1^2 - a_2^2 \end{pmatrix}$$

```
RHS1 = a*a' - norma2*eye(3)
```

RHS1 =

$$\begin{pmatrix} -a_2^2 - a_3^2 & a_1 a_2 & a_1 a_3 \\ a_1 a_2 & -a_1^2 - a_3^2 & a_2 a_3 \\ a_1 a_3 & a_2 a_3 & -a_1^2 - a_2^2 \end{pmatrix}$$

```
shouldbezero3times3 = LHS1 - RHS1
```

shouldbezero3times3 =

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

We now wish to prove eq. (2.13):

$$\widehat{a}^3 = -||a||^2\widehat{a}$$

In the code below,

$$-||a||^2\widehat{a}$$

is defined as RHS2, and $\widehat{a}^3$ as LHS2. The equation will be proven if

LHS2 - RHS2 = 0.

```
LHS2 = simplify(LHS1*hata)
```

LHS2 =

$$\begin{pmatrix} 0 & a_3\,\sigma_1 & -a_2\,\sigma_1 \\ -a_3\,\sigma_1 & 0 & a_1\,\sigma_1 \\ a_2\,\sigma_1 & -a_1\,\sigma_1 & 0 \end{pmatrix}$$

where

$$\sigma_1 = a_1^2 + a_2^2 + a_3^2$$

```
RHS2 = -norma2*hata
```

RHS2 =

$$\begin{pmatrix} 0 & a_3\,\sigma_1 & -a_2\,\sigma_1 \\ -a_3\,\sigma_1 & 0 & a_1\,\sigma_1 \\ a_2\,\sigma_1 & -a_1\,\sigma_1 & 0 \end{pmatrix}$$

where

$$\sigma_1 = a_1^2 + a_2^2 + a_3^2$$

```
shouldbezero3times3 = LHS2 - RHS2
```

shouldbezero3times3 =

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$